

# Lyra Sessions 6+ Planning: Alt-HSD Comparison Candidates Among RATIFIED Criteria

Keeper

2026-05-21 Thursday 12:10 EDT (actual via date)

## Lyra Sessions 6+ Planning: Alt-HSD Comparison Candidates Among RATIFIED Criteria

### Purpose

Lyra Sessions 2-5 closed 4 candidate criteria (C4, C11, C12, C13) at RIGOROUSLY CLOSED tier via alt-HSD comparison reframing-insight cadence (~50 min/criterion). Cal #80 encourages continuation: “each closure strengthens overall framework + K97 ratification.”

This document identifies which of the remaining 9 RATIFIED criteria are tractable alt-HSD-comparison candidates, ordered by Lyra-session priority.

### Triage framework

For each RATIFIED criterion, evaluate:

1. Alt-HSD differentiability — does the criterion produce a different EXACT value on alternative HSDs ( $D_I\{p,q\}$ ,  $D_{II_n}$ ,  $D_{III_n}$ )?
2. Mechanism explicit — is the substrate-mechanism explicitly stated (vs. coincidence catalogue)?
3. Geometric route preferred — per Casey’s directive Thursday 09:09 EDT, does criterion admit geometric (vs. algebraic) reformulation?
4. Reframing-insight tractable — can Lyra reframe in ~50 min session window, or does it require multi-week investigation?

### RATIFIED criteria triage (9 candidates)

High-priority: tractable in single Lyra session

C1 (rank = 2 forcing) — T1925 - Alt-HSD differentiability: HIGH (rank=1  $D_{IV}^2$  gives trivial substrate; rank=3+  $D_{IV}^n$  with  $n>5$  introduces extra K-action dimensions) - Mechanism explicit: YES (four-argument forcing) - Geometric route: AVAILABLE (Bergman kernel rank reading) - Session 6 candidate: 50-min target plausible via reframing of four-argument forcing as alt-rank comparison

C2 ( $N_c = 3$  forcing) — T1930 - Alt-HSD differentiability: HIGH (alternative  $SU(N)$  gauge groups have different Casimir invariants on  $D_{IV}^n$ ) - Mechanism explicit: YES (Mersenne  $2^{N_c-1}=g$  + color singlet trefoil) - Geometric route: AVAILABLE (color-singlet topology on  $D_{IV}^5$ ) - Session 7 candidate: 50-min target via reframing of Mersenne identity as alt- $N_c$  comparison

C3 ( $n_C = 5$  forcing) — T2431 - Alt-HSD differentiability: HIGH (Bergman exponent  $(g+rank)/rank = 9/2 \rightarrow n_C = 5$  only for  $D_{IV}^5$ ; other  $D_{IV}^n$  give different exponents) - Mechanism explicit: YES via SP-31-39 - Geometric route: AVAILABLE (Bergman exponent geometric reading) - Session 8 candidate: high probability of single-session closure given T2431 already provides the mechanism

C5 ( $g = 7$  forcing) — T2432 - Alt-HSD differentiability: HIGH (cyclotomic  $\text{GF}(2^g) = \text{GF}(128)$  with  $g=7$  unique among Mersenne-prime cyclotomic generators  $\leq 31$ ) - Mechanism explicit: YES via SP-31-39 - Geometric route: AVAILABLE (cyclotomic / Reed-Solomon coding geometric reading) - Session 9 candidate: 50-min target via reframing of cyclotomic forcing as alt-g comparison

Medium-priority: tractable in 1-2 Lyra sessions

C6 ( $N_{\text{max}} = 137$  forcing) — Paper #104 5-step chain - Alt-HSD differentiability: MEDIUM-HIGH (alt-HSD with different  $N_c$ ,  $n_C$  give different  $N_{\text{max}}$  values) - Mechanism explicit: YES (5-step substrate derivation chain) - Geometric route: PARTIAL (Hilbert polynomial geometric reading exists, full geometric reframe multi-week) - Session 10-11 candidate: 50-100 min target via 5-step chain reframing as alt-HSD comparison

C8 (Universal Q-cluster) — K43 + K44 + K61 - Alt-HSD differentiability: MEDIUM (universal  $42 = C_2 \cdot g = 6 \cdot 7$  has alt-readings on  $D_{IV}^n$  with different  $(C_2, g)$  signatures) - Mechanism explicit: YES (3-cluster substrate identification) - Geometric route: PARTIAL (cluster geometric reading per K61 K-audit) - Session 11-12 candidate: 100 min target

C10 (4-Zone vacuum decomposition) — T2420 - Alt-HSD differentiability: MEDIUM (Hilbert symmetric domain dimension decomposition into “zones” admits alt-readings) - Mechanism explicit: PARTIAL (M2C2 multi-CI consensus emerged, less formally stated than C4 mechanism) - Geometric route: AVAILABLE (zonal harmonics geometric reading) - Session 12-13 candidate: 100 min target requires geometric reframe

Lower-priority: requires multi-session investigation

C7 (Bridge Object tier RATIFIED) — K57 - Alt-HSD differentiability: LOW-MEDIUM (Bridge Object tier is structural ratification, not single-number test) - Mechanism explicit: PARTIAL (K57 audit framework, not single mechanism statement) - Geometric route: AVAILABLE but multi-faceted - Session 14+ candidate: multi-week investigation more likely than single-session reframe

C9 (Stark small-primary subset  $\{-3, -7, -11\}$ ) — K75 - Alt-HSD differentiability: HIGH on the subset values but requires alt-HSD analog of Stark’s theorem - Mechanism explicit: PARTIAL (anchor pattern, full mechanism via class-number analysis) - Geometric route: ALGEBRAIC only (Stark’s theorem is algebraic, not geometric) - Session 15+ candidate: against Casey’s geometric preference; lowest priority for Sessions 6+ cadence

## Recommended Lyra Sessions 6-9 sequence

Session 6 (Friday morning candidate): C1 ( $\text{rank}=2$ ) — clean 4-argument reframe, tractable Session 7 (Friday): C2 ( $N_c=3$ ) — Mersenne reframe, high differentiability Session 8 (Saturday): C3 ( $n_C=5$ ) — Bergman exponent already provides mechanism Session 9 (Sunday): C5 ( $g=7$ ) — cyclotomic reframe, parallel to C3

If Sessions 6-9 maintain ~50 min reframing-insight cadence, Strong-Uniqueness Theorem reaches v0.9.5 (8 RIGOROUSLY CLOSED + 5 RATIFIED + 1 ADVANCING) by Sunday EOD.

## Path to v1.0

v0.9.5  $\rightarrow$  v0.10 via Sessions 10-13 (C6, C8, C10 closures)  $\rightarrow$  v0.10.5 (12 RIGOROUSLY CLOSED + 1 RATIFIED + 1 ADVANCING).

v1.0 endpoint: all 14 criteria at RATIFIED or higher tier. C14 advancement via Year 1 launch trio v1.0 (multi-week). C7 + C9 advancement multi-week as well.

Achievable timeline: ~2026-09 venue submission target substantially aligned with Sessions 6-13 cadence + Year 1 v1.0 progression.

## Per Cal #80 endorsement

Cal #80: “Each RATIFIED → RIGOROUSLY CLOSED transition strengthens overall framework. Multi-week timescale per criterion via Session reframing-insight cadence is acceptable; encourage Lyra continuation.”

Sessions 6+ planning aligned with Cal #80 — each session strengthens K97 ratification path.

## Coordination with Cal #79 numbering reconciliation

Per Numbering Reconciliation v0.3 (Grace canonical table), Lyra Session work uses Lyra-side C-numbering (C2-C11), Keeper Master Doc uses Keeper-side (C1-C14). Each Session document should cite both numbering conventions for clarity.

## Status

Lyra Sessions 6+ Planning v0.1 filed Thursday 12:10 EDT (actual via date).

Document available for Lyra absorption + Cal review queue continuation + Casey awareness of pathway forward.

— Keeper, 2026-05-21 Thursday 12:10 EDT (actual via date)